| INCH-POUND |

MIL-F-7179G
14 APRIL 1993
SUPERSEDING
MIL-F-7179F
25 SEPTEMBER 1984

MILITARY SPECIFICATION

FINISHES, COATINGS, AND SEALANTS FOR THE PROTECTION OF AEROSPACE WEAPONS SYSTEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope This specification establishes the minimum requirements for finishes, protective coatings, and sealants used in the protection of aerospace weapons systems (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

TT-L-190	-	Linseed Oil, Boiled, (For Use in Organic Coatings)
TT-P-28	-	Paint, Aluminum, Heat Resisting (1200°F)
TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
TT-P-2756	-	Polyurethane Coating: Self-Priming Topcoat, Low Volatile Organic Compounds (VOC)
TT-P-2760	-	Primer Coating: Polyurethane, Elastomeric
VV-L-800	-	Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Systems Requirements Department, Code SR3, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited
Source: https://assist.dla.mil -- Downloaded: 202610-17T08:15Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-R-3043 - Resin-Coatings, Unpigmented, For Engine Components and Metal Parts
MIL-S-5002 - Surface Treatments and Inorganic Coatings For Metal Surfaces of Systems
MIL-C-5056 - Coating, Permanent Resin; Process For Application of, to Aircraft Parts
MIL-B-5087 - Bonding, Electrical, and Lighting Protection, For Aerospace Systems
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-C-6529 - Corrosion Preventive, Aircraft Engine
MIL-C-8507 - Coating, Wash Primer (Pretreatment), For Metals, Application of (For Aeronautical Use)
MIL-C-8514 - Coating Compound, Metal Pretreatment, Resin-Acid
MIL-A-8625 - Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-C-8779 - Colors, Interior, Aircraft, Requirements for
MIL-S-8802 - Sealing Compound, Temperature-Resistant, Integral Fuel Tanks and Fuel Cell Cavities, High-Adhesion
MIL-A-9962 - Abrasive Mats, Nonwoven, Monmetallic
MIL-C-11796 - Corrosion Preventive Compound, Petrolatum, Hot Application
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold Application
MIL-F-18264 - Finishes, Organic, Weapons System, Application and Control of
MIL-C-22750 - Coating, Epoxy, VOC - Compliant
MIL-P-23377 - Primer Coatings: Epoxy, Chemical and Solvent Resistant
MIL-L-23398 - Lubricant, Solid Film, Air-Cured, Corrosion Inhibiting, NATO Code Number S-749
MIL-C-27725 - Coatings, Corrosion Preventive, For Aircraft Integral Fuel Tanks
MIL-S-29574 - Sealing Compound, Polythioether, For Aircraft Structures, Fuel and High Temperature Resistant, Fast Curing at Ambient and Low Temperatures
MIL-S-38249 - Sealing Compound, Firewall
MIL-L-46010 - Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting
MIL-C-46081 - Coating Compound, Thermal Insulating (Intumescent)
MIL-A-46146 - Adhesive-Sealants, Silicone, RTV, Non-corrosive (For Use With Sensitive Metals and Equipment)
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-P-53022 - Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free
MIL-P-53030 - Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free
MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-C-53072 - Chemical Agent Resistant Coating (Carc) System Application Procedures and Quality Control Inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

MIL-L-63460 - Lubricant, Cleaner and Preservative for Weapons and Weapons Systems
MIL-C-81309 - Corrosion Preventive Compounds, Water Displacing, Ultra Thin Film
MIL-S-81733 - Sealing and Coating Compound, Corrosion Inhibitive
MIL-C-83231 - Coating Polyurethane, Rain Erosion Resistant, For Exterior Aircraft and Missile Plastic Parts
MIL-C-83286 - Coating, Urethane, Aliphatic, Isocyanate, For Aerospace Applications
MIL-A-83377 - Adhesive Bonding (Structural) for Aerospace and Other Systems, Requirements for
MIL-S-83430 - Sealing Compound, Integral Fuel Tank and Fuel Cell Cavities, Intermittent Use to 360°F (182°C)
MIL-C-83445 - Coating System, Polyurethane, Non Yellowing, White, Rain Erosion Resistant, Thermally Reflective
MIL-T-83483 - Thread Compound, Antiseize, Molybdenum Disulfide Petrolatum
MIL-C-85054 - Corrosion Preventive Compound, Water Displacing Clear (Amlguard)
MIL-C-85285 - Coating Polyurethane, High-Solids
MIL-C-85322 - Coating, Elastomeric, Polyurethane, Rain-Erosion
MIL-P-85582 - Primer Coatings: Epoxy, Waterborne

STANDARDS

MILITARY

MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-889 - Dissimilar Metals
MIL-STD-970 - Order of Preference for the Selection of Standards and Specifications
MIL-STD-2161 - Paint Schemes and Exterior Markings for U.S Navy and Marine Corps Aircraft

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 202640-17T08:15Z
Check the source to verify that this is the currentversion before use.

3 REQUIREMENTS

3.1 Instructions to contractors

3.1.1 General. It shall be the responsibility of the contractor to ensure that areas of the aerospace weapons system (see 6.2) exposed to extreme conditions are protected by coatings and finishes proven suitable for the intended environment and must be approved by the procuring activity. Examples of extreme conditions include, but are not limited to exhaust trails, gun-blast surfaces, rocket-blast areas, hull bottoms (see 6.2), leading edges, areas that may trap or be exposed to fumes from exhaust, guns, or rockets, or surfaces subjected to temperatures above 250°F (121°C) as the result of thermal radiation, aerodynamic heating, or other sources of heat.

3.1.1.1 Non-standard coatings and finishes. Surfaces and areas that may require protection with a non-standard coating or finishing system shall be protected by coatings and finishes that have been proven suitable for the intended environment and approved by the procuring activity.

3.1.1.2 Exceptions. Unless specifically prescribed by a particular item and/or accessory specification or drawing, this specification does not apply to the following: propellers, powerplants, instruments, electronic enclosures or chassis, or standard contractor purchased accessories, such as electric motors, generators, switches, hydraulic valves and similar parts. Contractors are not required to refinish such equipment or accessories, provided that the color and finish system thereon are acceptable to the contractor and the procuring activity or its agent. However, all surfaces in contact with parts covered herein shall be protected in accordance with the requirements of this specification.

3.1.2 Contractor finish specification. A specification defining the protective finish system, including cleaning, surface treatment to be applied to the contractual articles shall be developed by the contractor and submitted for approval by the procuring activity. The contractor finish specification shall use MIL-F-7179 as its basis for preparation and format. Any proposed deviations from the requirements of specified herein shall be supported by applicable technical data and information (see 3.1.2.2).

3.1.2.1 Special treatments. A detailed parts list of the weapon system is not required in the contractor finish specification, however, sufficient examples shall be included in the finish specification so that any part of the weapon system may be properly classified as to its detailed protective treatment. All parts for which the exact treatment may be in doubt, all of the areas listed in 3.1.1 requiring special treatment, and all parts or assemblies with design features noted in their specifications to require maximum corrosion preventive treatment, shall be included. When the contractor finish specification references manufacturer's material, coating, and/or finish process specifications, or other documents, there shall be a cross-reference supplied as part of the specification indicating their equivalence to existing Government and non-Government specifications or standards. Copies of all referenced contractor originated documents shall be submitted with the proposed coating or finish specification.

Source: https://assist.dla.mil -- Downloaded: 202640-17T08:15Z
Check the source to verify that this is the currentversion before use.

3.1 2 2 Alternate materials. Where there is a choice of coatings and/or finish materials or processes permitted herein, the contractor finish specification shall list only those intended to be used Any proposed deviations from, or additions to, the requirements specified herein shall be marked with an asterisk in the contractor finish specification and shall be supported by applicable technical test data and information to permit a thorough engineering evaluation. It shall be the responsibility of the prime contractor to ensure that sub-contractors meet all of the requirements of the prime contractor finish specification and the applicable requirements of this specification When subcontractors propose the use of alternate materials or processes, the proposals shall be specifically reviewed and approved by the procuring activity and the prime contractor prior to use Additionally, these materials shall comply with all applicable federal, state, and local environmental and occupational safety/hazard laws and regulations.

3.1 3 Selection of specifications and standards The order of preference for the selection of specifications and standards for the development of the contractor finish specification shall be in accordance with MIL-STD-970

3.2 Materials and processes. All coating and finish materials used shall conform to the requirements of the documents referenced in this document. Alternate materials shall not be used without the written approval by the procuring activity.

3.3 Work done prior to material inspection. Any coating, finish, or sealing work done prior to the material inspection shall be at the risk of the contractor. If the materials fail to meet the applicable specification requirements in any respect, the lot of material shall be rejected. Any coating and sealing work accomplished using rejected material shall require stripping of the rejected material and application of approved material.

3.4 General protective finish system requirements.

3.4.1 Surface treatment. All metal surfaces, regardless of whether they are to be painted or are specifically excluded from painting requirements, shall be surface treated in accordance with MIL-S-5002, except as modified by 3.6.2.1c.

3.4.2 Application of organic finishes. The application of organic coatings and finish systems prescribed herein shall be in accordance with MIL-F-18264 or MIL-C-53072, as applicable

3.4.3 Interior surfaces. Primer coating and topcoat, where applicable, as specified in tables I and II, shall be applied to the interior surfaces of items in accordance with MIL-F-18264 or MIL-C-53072, as applicable. The primer coating shall be applied such that the dry-film thickness is in accordance with 3.4.5 and tables I and II, with the exception that the topcoat may be applied after final assembly, subject to the requirements of 3.6.2. When a topcoat is required for interior surfaces, such as to prevent fluid intrusion or to enhance visibility, and those surfaces are primed and topcoated prior to assembly, a finish coat shall be applied after final assembly.

Source: https://assist.dla.mil -- Downloaded: 202640-17T08:15Z
Check the source to verify that this is the currentversion before use.

3.4.4 Exterior surfaces Primer coatings and topcoats shall be applied to the exterior surfaces (see 6.2) of items, as specified in 3.4 5 and tables I and II, and in accordance with MIL-F-18264 or MIL-C-53072, as applicable

3.4.5 Coating thickness. The minimum applied dry-film thickness of the coatings in table I shall be as specified in table II. On interior surfaces of all materials and on the exterior surfaces of magnesium the applied dry-film thickness of the coatings shall not exceed that specified in table II by more than 50 percent. The maximum applied dry-film thickness of the coatings in table I are to be as follows:

Coating	Maximum applied dry-film thickness (mils (μm))
MIL-P-23377	0 9 (23)
MIL-P-53022	1.5 (38)
MIL-P-53030	1.5 (38)
MIL-P-85582	1.2 (30)
TT-P-2760	2.0 (51)
MIL-C-22750	2.3 (58)
MIL-C-46168	1/
MIL-C-53039	1/
MIL-C-83286	2.3 (58)
MIL-C-85285	2.3 (58)
TT-P-2756	2.3 (58)

1/ See MIL-C-53072

3.4.5.1 Coating thickness deviations. In areas of application where the required coating thickness cannot be met, the contractor shall indicate in the finish specification (see 3.1.2) specific areas of deviation.

Source: https://assist.dla.mil -- Downloaded: 202610-17T08:15Z
Check the source to verify that this is the current version before use.

3.5 General precautions

3.5.1 Metal particles. Fabrication and assembly procedures shall be established such that the retention of metal particles or pieces, such as chips, slivers, and filings, or other foreign objects in structures is precluded. A vacuum cleaner providing strong suction shall be employed for frequent cleaning operations in relatively inaccessible areas. Metal cutting or filing is not permitted on an assembly after it has been accepted, except upon specific approval of the local government representative, when such action has been approved, those areas shall be retouched in accordance with the detailed paint schedule for the part.

3.5.2 Use of abrasive materials. The selection of the correct abrasive material and size is dependent upon the material/surface to be treated and the purpose for which the abrasive materials is to be used In areas where control or removal of generated debris will cause failure of the weapon system, the abrasive shall not be used Applicable technical orders, bulletins, and manuals shall be consulted prior to selection of any abrasive material Any abrasive material that may produce a surface finish that would degrade the fatigue life of the part shall not be used. Abrasives used on one metal or alloy shall not be used on different metals or alloys unless it can be proven to the procuring activity that the media recycling process eliminates damaging contaminants

3.5.2.1 Metallic wool. Metallic wools selected for use on a particular material/surface shall be galvanically compatible with the material/surface. Steel wool may be used on low-alloy steel. Aluminum wool may be used on aluminum, aluminum alloys, magnesium, and magnesium alloys. Copper wool may be used on copper alloys, bronze, and brass. Stainless steel wool may be used on stainless steel. After metallic wools are used on a material/surface, all residue must be removed from that material/surface. Any residue not removed is a potential source for galvanic corrosion.

3.5.2.2 Wire brushes. Wire brushes shall not be used on those parts where the resultant surface finish would degrade the fatigue life of the part. Wire brushes selected for use on a particular material shall be galvanically compatible with the material. Only austenitic stainless steel brushes may be used on austenitic stainless steel surfaces. When an austenitic stainless steel brush is used, it shall be dedicated to only one given material to avoid cross-contamination. Use of brushes with bristle wire diameters greater than 0.01 in. (0.25 mm) is prohibited. After use of a wire brush on soft metals, such as aluminum or magnesium, the surface must be polished with fine abrasive paper or abrasive mat conforming to MIL-A-9962, Type I, Grade A or B.

Source: https://assist.dla.mil -- Downloaded: 202610-17T08:15Z
Check the source to verify that this is the currenversion before use.

3 5 2.3 General wire brush selection The following may be used as a general compatibility guide in selecting wire brushes for use on selected metal alloys. This listing is not all-inclusive and does not give blanket authorization for the use of wire brushes on equipment and materials. When the use of wire brushes on sections of equipment is intended, it must be stated in the contractor finish specification (see 3 1.2).

Alloy construction	Wire brush construction			
	Stainless Steel	Aluminum	Carbon Steel	Brass
Aluminum	X	X		
Magnesium	X	X		
Ferrous (Non-stainless)	X		X	
Stainless steel and Nickel	X			
Copper				X
Titanium	X			
Cadmium- or zinc-plated 1/				
Chromium-, Nickel-, Tin-, or Copper-plated	X			

1/ The use of wire brushes on cadmium-plated and zinc-plated alloys is not recommended.

3.5.3 Welding, brazing, and soldering. When welding, brazing, or soldering is permitted on an assembly after it has been finished, inspected, and accepted, the surface shall be refinished. Any pre- or post-cleaning processes associated with welding, brazing, or soldering operation shall be compatible with subsequent finish processes and shall be compatible with the joining materials (filler materials, solder, fluxes, or brazing alloys). Any surface refinishing shall cover a large enough area to ensure correction of any damage resulting from the joining operation. The finish requirements in the joined area shall be equivalent to the basic part finish requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

3.6 Requirements applicable to specific parts

3.6.1 General. In applying the requirements of this section, the groupings of the more commonly used aircraft metals shall be selected in accordance with MIL-STD-889, with the exception that the protection requirements specified for attaching parts and fasteners (see 3.6.13) shall take precedence, where applicable. Metal foils or shims shall not be used on exterior surfaces without suitable protection.

3.6.2 Fayed surfaces, joints, and seams.

3.6.2.1 Surfaces of similar metals. Seams and joints that possess fayed surfaces (see 6.2) of similar metals, as defined by MIL-STD-889, shall be protected, at a minimum, by application to each surface of primer coating in accordance with 3.4.5 and tables I and II. The dry film thickness of the primer coating shall be as prescribed for interior surfaces (see 3 4 3, 3 4 5 and tables I and II) Exceptions to the above are as follows.

a Where 3.4 5 and tables I and II specify application of a specific thickness of primer coating to fayed surfaces, one-half of the required thickness of primer coating may be applied to each surface being joined.

b. Primer coating shall not be applied to resistance-welded fayed surfaces. Only weld-through sealants approved by the procuring activity shall be used prior to assembly. Primer coating shall be applied to fayed surfaces to the extent practicable after spot-welding. All exterior edges shall be sealed.

c. Fayed surfaces that are to be adhesively bonded shall be surface cleaned, treated, and processed as specified in the procuring activity approved bonding procedures document(s) for the assemblies concerned, or in accordance with MIL-A-83377, as applicable.

d. Titanium to titanium and corrosion resistant steel to corrosion resistant steel constructions shall be protected by application of primer coating (see 3.4.5 and tables I and II) or sealant, conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430, to the fayed surfaces. Where protection against fretting is required for these constructions, the contractor shall propose a method of protection for approval by the procuring activity.

e. In addition to any required primer coating, all exterior fayed surfaces, seams, and edges shall be sealed with a sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430. A minimum gap of 0.02 in. (0.5 mm) is required at exterior surface (see 6.2.2) butt joints to allow for effective sealing.

3.6.2.1.1 Defect filling. The use of filling material for the purpose of sealing and concealing nicks, dents, gouges, and poor joints is prohibited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

3.6 2.2 Surfaces of dissimilar metals. Surfaces of dissimilar metals, as defined in MIL-STD-889, shall each receive a minimum of 0 6 mil (15 µm) of primer coating in accordance with table I, except as specified in 3.6.2.1c. When fayed surfaces are of dissimilar metals, they shall be sealed with sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430, and the thickness of the primer coating applied to each surface shall be in accordance with 3 4 5 and table II. In addition, the following precautions shall be taken:

a. Where magnesium is part of a dissimilar metal fayed surface, sealant conforming to MIL-S-29574, Type II, or MIL-S-81733 shall be applied between the surfaces and squeezed out of all boundaries. The excess shall be removed in a manner that will ensure a fillet on all edges Except for bushing installations, the fillet width shall be a minimum of 0.25 in (6.4 mm) For bushings, the fillet shall be the largest practicable Joint areas which may retain water shall be filled with sealing compound.

b Butt joints in exterior locations consisting of dissimilar metals shall be protected by grooving a seam to a width of 0.09 ± 0.03 in. (2.3 ± 0 76 mm) and filling with sealing compound. The depth of the groove shall be of sufficient depth so as to retain the hardening type sealing compound, which shall be subsequently applied and smoothed flush with the surfaces of adjacent dissimilar metals.

c. In joints constructed of a carbon-fiber reinforced composite and aluminum, or other dissimilar metal (as defined by MIL-STD-889), there shall be a final ply of material composed of the same resin matrix but reinforced with a material galvanically compatible with the aluminum. The final ply shall extend a minimum of 0.25 in. (6.4 mm) beyond the metal member. Primer coating shall be applied to a dry-film thickness of 1.2 to 1.8 mils (30 to 46 µm) to each of the interfacial surfaces. The surfaces of permanent joints shall be fayed and sealed with sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430. Joints that require separation as part of normal maintenance may have formed-in-place seals substituted for fayed surface sealing.

3.6.3 Sealing. For exterior locations (see 6.2.2), openings (with the exception of drain holes at low points) which are not required for aircraft operations shall be sealed to prevent fluid intrusion from external sources. Sealing around access plates shall be accomplished by the application of sealant to the structure in a manner such that the access plates can be removed without damaging the formed-in-place sealant or the surrounding metal. The minimum thickness of sealant for formed-in-place seals shall be 0.030 in (0 76 mm)

Source: https://assist.dla.mil -- Downloaded: 202610-17T08:15Z
Check the source to verify that this is the current version before use.

3 6.4 Slip fits. The sealing of slip fits shall be accomplished with wet primer coating conforming to MIL-P-23377, Type I or II, Class 1 or 2, TT-P-1757, or wet sealant conforming to MIL-S-29574, Type II, or MIL-S-81733. If design requires disassembly, primer coating conforming to MIL-P-23377, Type I or II, Class 1 or 2, or TT-P-1757 shall be applied and permitted to dry thoroughly prior to assembly. In instances where the above materials are incompatible with the function of the part or assembly, corrosion preventive compound conforming to MIL-C-16173, Grade 3 or 4, shall be used.

3.6 5 Press fits. The sealing of press fit component assemblies, with the exception of assemblies permanently housed in grease or oil, shall be accomplished with either wet primer coating conforming to MIL-P-23377, Type I or II, Class 1 or 2, TT-P-1757, or wet sealant conforming to MIL-S-81733 or MIL-S-29574, Type II. Exterior edges of the press fit component shall be sealed with sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430, with the exception that sealing with primer coatings (see 3 4 5 and table II) may be used for bushings with walls of 0.094 in. (2 4 mm) or less. The completed assembly shall then be finished as specified in 3 4 5 and table II. Parts permanently housed in grease or oil shall be assembled with the grease or oil to be used in the housing.

3.6.6 Cut edges. The edges of all metals shall be rounded to permit adhesion of an adequate thickness of applied paint coatings or sealants. After rounding of the edges, and prior to the application of paint or sealant, applicable chemical surface treatments shall be applied.

3.6.7 Functional surfaces. Paint-type coatings shall not be applied to functional, working, or wearing surfaces; to lubricated surfaces; to adjustable screw threads; to lubrication or drain holes; to bearing or sliding surfaces; to areas where they could be rubbed or scraped onto surfaces which must be clean and bare to function properly; or to any other surface where the application of the coating may cause malfunction of the part or system. The interior walls of drain holes shall be coated with paint-type coatings for corrosion protection.

3.6.8 Control cables and control chains. Control cables and control chains shall not be painted. However, prior to installation, control cables shall be protected with a dip-coating of corrosion preventive material conforming to MIL-C-16173, Grade 4, with the exception of those surfaces requiring lubrication for functional purposes. Those surfaces requiring lubrication shall be cleaned and coated with the required lubricant in lieu of corrosion preventive material. After installation, the control cables and control chains shall be inspected. If touch-up of the corrosion preventive compound is necessary, touch up shall be accomplished with the same material used prior to installation. Nylon jacketed cables do not require treatment, with the exception of exposed end fittings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

3 6 9 Closely coiled springs. Springs that are closely coiled, preventing the application of plating to internal surfaces, or springs not plated for other reasons, shall receive a minimum of 1.2 mils (31 µm) of primer coating conforming to MIL-P-23377 or MIL-P-85582, or shall be coated with corrosion preventive compound conforming to MIL-C-16173, Grade 4, or MIL-C-11796, Class 2.

3.6.10 Parts in oil or grease. Parts which are housed in lubricating oil, hydraulic oil, or grease shall finished with a baked resin coating conforming to MIL-R-3043, applied in accordance with MIL-C-5056. Parts constructed of corrosion-resistant metals need not be coated, unless they contact dissimilar metals Functional surfaces, such as bearing surfaces, shall not be coated.

3.6.11 Metal tanks

3.6.11 1 Temporary and auxiliary fuel tanks. The inside surfaces of aluminum alloy tanks shall be surface treated in accordance with MIL-C-5541, Class 1A, or MIL-A-8625, Type II, but shall not be painted The interior of steel tanks shall be finished with baked resin coating conforming to MIL-R-3043, applied in accordance with MIL-C-5056. Sealant conforming to MIL-S-83430 or MIL-S-29574, Type II, may be used in lieu of MIL-R-3043 when authorized by the procuring activity.

3.6.11.2 Welded fuel tanks (including external auxiliary fuel tanks). The interior surfaces of aluminum tanks shall be thoroughly cleaned, and surface treated in accordance with MIL-C-5541, Class 1A, sloshed with corrosion preventive compound conforming to MIL-C-6529, Type II, and drained, but shall not be painted. Droppable steel tanks shall be finished on the interior with a baked resin finish conforming to MIL-R-3043, applied in accordance with MIL-C-5056, or sealant conforming to MIL-S-83430 or MIL-S-29574, Type II.

3.6.11.3 Integral and riveted fuel tanks. The inside surface of integral or permanently fastened, such as riveted, fuel tanks shall be suitably finished and sealed with sealant conforming to MIL-S-8802, MIL-S-29574, or MIL-S-83430 to prevent corrosion and leakage of fuel. Interior surfaces that are fayed, whether sealed or not, shall be coated with 0.9 to 1.5 mils (23 to 38 µm) of corrosion preventive coating conforming to MIL-C-27725. MIL-C-27725 shall not be applied to exterior surfaces.

3.6.11.4 Lubricating oil and hydraulic fluid tanks. The inside surfaces of lubricating oil tanks constructed of corrosion-resistant materials shall not be painted. Tanks of other materials shall be finished with a baked resin coating conforming to MIL-R-3043, applied in accordance with MIL-C-5056.

3.6 11 5 Miscellaneous aluminum alloy tanks. Interior surfaces of miscellaneous aluminum alloy tanks, with the exception of potable water tanks, shall be surface treated in accordance with MIL-A-8625, Type II, or MIL-C-5541, Class 1A. The interior surfaces of potable water tanks shall not be painted or conversion coated.

Source: https://assist.dla.mil -- Downloaded: 202640-17T08:15Z
Check the source to verify that this is the current version before use.

3 6 12 Tubing.

3.6.12.1 Nonstructural tubing. With the exception of tubing constructed of titanium alloy, corrosion-resistant steel alloy, heat resistant steel alloy, and as otherwise specified herein, all nonstructural tubing and plumbing lines shall receive the complete interior or exterior paint system, as applicable, on the exterior of the lines, and shall be protected in accordance with table III.

3 6.12.1.1 Oxygen tubing. Surface finishes, such as paints, primer coatings, electrical coatings, and conversion treatments, such as anodizing or non-electrochemical chromate, or phosphate conversion coatings, are not to be applied to the interior of oxygen tubing. Oxygen tubing is to be thoroughly cleaned of all contaminants prior to any mechanical processing, such as double boring, being performed.

3 6 12.1.2 Aluminum tubing and plumbing lines. Interior and exterior surfaces of aluminum-alloy tubing and plumbing lines shall be surface-treated in accordance with MIL-C-5541, Class 1A, or MIL-A-8625, Type II The exterior of aluminum plumbing lines in fuel tanks require the application of a corrosion preventive coating conforming to MIL-C-27725 Paint coatings shall not be applied to the interior surfaces of airspeed indicator tubing or other sensing lines. Aluminum tubing used in fire-extinguishing systems employing halogenated agents shall be finished internally and externally with a baked resin coating conforming to MIL-R-3043, applied in accordance with MIL-C-5056

3.6.12.1.3 Protection of tubing joints after installation. After installation of the tube assemblies, all remaining non-sealed joints, which will not be disconnected during normal servicing, shall receive a coating of sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430, followed by the appropriate paint system. All remaining nonsealed joints which must be disconnected during normal servicing operations shall be coated with corrosion preventive compound conforming to MIL-C-16173, Grade 4, or MIL-C-85054, which shall seal all exposed spaces between the parts. A second coat of the same material shall be applied to the same areas after a period of 60 minutes. Contractor-prepared maintenance instructions shall require periodic reapplication of this material in-service.

3.6.12.2 Structural tubing.

3.6.12.2 1 Structural magnesium tubing. The use of magnesium or magnesium alloy is prohibited for structural tubing.

Source: https://assist.dla.mil -- Downloaded: 202640-17T08:15Z
Check the source to verify that this is the currentversion before use.

3.6 12.2 2 Structural carbon steel tubing. All exterior surfaces and all interior surfaces without completely welded or crimped ends of structural carbon steel tubular assemblies shall be finished in accordance with 3 4 5, tables I and II, and with the following, where applicable

a. Assemblies completely closed by welding or to which the application of primer coating is impractical or ineffective, such as crimped-end tubing not closed by welding or tubing heat treated after assembly, shall be treated after assembly (and heat treated, if necessary) with hot (160°F (71°C)) linseed oil conforming to TT-L-190, or corrosion preventive compound conforming to MIL-C-11796, Class 1 or 1A, or MIL-C-16173, Grade 2 or 4. The corrosion preventive compound shall be applied under pressure into the hollow member through holes drilled in the tubing, or by immersion of the tubing in a bath of the preservative.

b. For large tubing structures, interconnecting holes may be drilled between members to promote circulation of the corrosion preventive compound, described in 3 6 12 2a

c. Parts subjected to immersion shall be manipulated in such manner to ensure the absence of air pockets and shall remain in the bath until all bubbling has ceased. The members shall be thoroughly drained after treatment and all access holes drilled in the members shall be closed with cadmium plated, self-tapping screws or blind rivets. The screws or rivets shall be wet installed with sealant conforming to MIL-S-81733 or MIL-S-29574, type II, and overcoated with the same after installation The exterior surface of the tubing assemblies shall be free of oil, grease, and dirt prior to application of the prescribed finish system.

3.6.12.2.3 Structural aluminum-alloy tubing. Interior surfaces of structural aluminum-alloy tubing shall be protected in accordance with 3.4.5 and tables I and II. The interior surfaces of structural aluminum-alloy tubing closed by welding shall be coated with primer coating conforming to MIL-P-23377, Type I or II, Class 1 or 2, or corrosion preventive compound conforming to MIL-C-16173, Grade 2 or 4, applied through appropriately drilled holes.

3.6.12.2.4 Structural copper alloy, corrosion-resistant alloy, and heat-resistant alloy tubing. Interior and exterior surfaces of structural copper alloy, corrosion-resistant alloy, and heat-resistant alloy tubing need not be painted, except as required for dissimilar metal contact.

3.6.12.3 Mechanical attachment. Tubular parts which have fittings mechanically attached shall have all edges of the attachment sealed with a sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430.

Source: https://assist.dla.mil -- Downloaded: 202610-17T08:15Z
Check the source to verify that this is the current version before use.

3 6 13 Attaching parts and fasteners

3.6.13.1 General finish. Attaching parts and fasteners, such as screws, nuts, bolts, bushings, spacers, washers, rivets, high-shear rivets, self-tapping screws, sleeves for "shakeproof" fastener studs, self-locking nuts, "speed nuts", and clamps, need not be painted in detail, except when dissimilar metals are involved in the materials being joined. All attaching parts, or the surfaces with which they are in contact, however, shall be wet installed with primer coating conforming to MIL-P-23377, Type I or II, Class 1 or 2, TT-P-1757 or sealant conforming to MIL-S-81733 or MIL-S-29574, Type II. Primer coating or sealant shall not be applied to the threaded portions of fasteners for which torque requirements are established without the coating. When installed in aluminum structures, all steel, cadmium plated, and non-aluminum fasteners shall be overcoated with sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430 Dry-film thickness of sealant shall be a minimum of 6 mils (152 µm). For magnesium dissimilar metal combinations, the general requirements of 3.6.2 2 shall apply.

3 6 13.2 Close tolerance bolts. Prior to installation, close tolerance bolts shall receive application of primer coating conforming to MIL-P-23377, Type I or II, Class 1 or 2, TT-P-1757, sealant conforming to MIL-S-81733, except where frequent removal of the bolts is required. For close tolerance bolts requiring frequent removal, corrosion preventive compound conforming to MIL-C-11796, class 3, or MIL-C-16173, class 3, or corrosion-inhibiting, solid film lubricant conforming to MIL-L-46010 or MIL-L-23398. MIL-L-46010 requires heat curing and shall not be used on aluminum parts. MIL-L-23398 is air curing and may be used on all types of metallic parts. When a solid film lubricant is used, it shall be applied and completely cured prior to assembly. The bolt shall then be wet installed and fillet sealed (after installation) with sealant conforming to MIL-S-81733

3.6.13.3 Adjustable parts. Threads of adjustable parts, such as tie rods and turnbuckles, shall be lubricated and protected, both before and after assembly, with anti-seize compound conforming to MIL-T-83483, with lubricating oil conforming to MIL-L-63460, with corrosion preventive compound conforming to MIL-C-16173, Grade 3, or with lubricating oil conforming to VV-L-800 followed by corrosion preventive compound conforming to MIL-C-16173, Grade 2.

3.6.13.4 Touch-up. All attaching parts shall receive final finishing after installation. Topcoats shall be applied over the primer coating to match the color of adjacent exterior surfaces, when necessary. Nuts and heads of bolts in joints that are subsequently lubricated need not receive final finishing.

3.6.13.5 Washers. Washers constructed of aluminum alloy 5356 or 5052, or high pressure phenolic laminates of suitable design shall be used under machine screws, countersunk fasteners, bolt heads and nuts that would otherwise contact magnesium These parts and washers shall be wet installed and fillet sealed after installation with sealant conforming to MIL-S-81733

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

3 6 14 Areas subjected to corrosive fluids Battery compartments constructed of leakproof and corrosion resistant material require no further finishing. All other battery compartments and adjacent areas subject to vapors and spills shall be coated with a suitable polyurethane casting resin approved by the procuring activity. All bilge areas, all surfaces within 24.0 in. (610 mm) of urinals, and all areas beneath lavatories and galleys shall be finished with a minimum of 2 0 mils (51 µm) of primer coating conforming to TT-P-2760, or with a minimum of 0.9 mils (23 µm) of MIL-P-23377 or MIL-P-85582 and topcoated with coating conforming to MIL-C-83286 or MIL-C-85285, in accordance with 3.4.5 and tables I and II for exterior surfaces.

3.6 15 Fastenings and strut ends on seaplanes. All fastenings, strut ends, and other similar parts of seaplanes (see 6.2) exposed to the action of sea water or salt spray shall receive additional protection in the form of a coat of corrosion preventive compound conforming to MIL-C-16173, Grade 4 Subsequent to painting, all open-ended struts shall be coated by dipping in corrosion preventive compound conforming to MIL-C-16173, Grade 4, followed by draining and wiping the exterior surfaces prior to installation If it is not possible to coat parts completely by dipping, application by brush or spray is permissible

3.6.16 Float bumpers. The forward face of the float or hull under the bumper pad and all parts of the bumper shall receive a coat of corrosion preventive compound conforming to MIL-C-16173, Grade 4, in addition to the protection required by 3.4.5 and tables I and II.

3 6.17 Electrical parts.

3 6.17.1 Connections. The exterior of electrical bonding and ground connections conforming to MIL-B-5087 shall be finished in accordance with 3.4 5 and table II after installation. All permanent electrical bonds, such as jumpers and ground studs, shall be sealed after installation with sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430. Connectors not permanently sealed shall be internally and externally protected with material conforming to MIL-C-81309, Type III.

3.6.17.2 Conduit and boxes. Electrical conduit and junction or relay boxes shall receive protection in accordance with 3.4.5 and tables I and II. Plastic coated and braided wire shall not be coated.

3.6.17.3 Electrical connector pins and sockets. Electrical connector pins and sockets shall have a thin, continuous coat of MIL-C-81309, type III, applied to the pins and/or pin receptacle end of the connector prior to mating the connector halves. The connector shells shall then be coated with MIL-C-81309, type III after mating the connector halves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

3.6.18 Surfaces and components exposed to high temperatures. Areas and components that are exposed to temperature ranges.

a. 300° to 400°F (149 to 204°C), either on the ground or in flight (other than instantaneous effects), shall be finished in accordance with 3.4.5 and tables I and II in the appropriate color and gloss.

b. 400° to 500°F (204° to 260°C), a silicone finishing system shall be applied directly to surface-treated metal, omitting the wash primer and primer coating. The color shall conform to the color scheme for the weapons system. For exposures up to 450°F (232°C), sealant conforming to MIL-A-46146 may be used, when authorized by the procuring activity.

c. Above 500°F (260°C), heat-resistant finishes conforming to TT-P-28 may be used, however, each application must be approved by the procuring activity.

d For exposure to operational temperatures of 250° to 350°F (121° to 177°C), sealant conforming to MIL-S-38249, MIL-S-83430 shall be used. MIL-S-29574 may be used in areas where the operational temperature is a maximum of 300°F (149°C)

3.6.18.1 Fire insulating paint for Naval aircraft. Within power-plant compartments of Naval aircraft and other compartments normally operating at temperatures below 300°F (149°C), where fires are likely to occur as a result of flammable fluid leakage, and in areas adjacent to bleed air ducts and valves, which contain air at temperatures above 300°F (149°C), all fluid containers, such as air bottles, oxygen containers, hydraulic reservoirs, accumulators, and cylinders which could escalate intensity of a fire by explosion due to excessive heat, shall be protected by a finish system consisting of a minimum of 0.6 mils (15 μm) of primer coating conforming to MIL-P-23377 or MIL-P-85582 plus a minimum of 0.020 in. (0.51 mm) dry film thickness of MIL-C-46081 thermal insulating paint (normal interior finish requirements apply).

3.6.18.2 Coatings for temperature control. Surfaces subject to heating due to radiation from adjacent hot components or from exposure to a thermal pulse shall be finished with low-absorption coatings. Each proposed use of coatings for control of temperature shall be submitted as part of the contractor finish specification (see 3.1.2) to the procuring activity for approval. The request for approval shall include all necessary technical information concerning the proposed material and application, with data supporting the effectiveness of the coating system.

3.6.19 Hull and float bottoms. Flying-boat hull bottoms and float bottoms shall be finished with a system in accordance with tables I and II that provides protection from the erosive effects of high water speeds, and shall be aerodynamically smooth. Use of rubber grommets under the head of rivets, bolts, and screws on the exterior skin is prohibited. Where antifouling paint is prescribed, the finish shall be approved in each instance by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

3.6.20 Wood and phenolic surfaces. Wood and phenolic surfaces shall be finished with a minimum of two coats of varnish or enamel, plus an additional two coats, if in contact with metal surfaces or in exterior locations, in accordance with MIL-STD-171.

3.6.21 Molded plastic and ceramics. Transparent plastic parts shall not be painted. Other plastic parts (except fiberglass laminates, antenna and magnetic airborne detector (MAD) housing, and radomes) need not be painted, except for color matching purposes. Plastic or ceramic insulators (used for radio antennae) shall not be painted. Their edges, however, shall be sealed with sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430, after installation in exterior locations.

3.6.22 Finishing of ducts. The interior surfaces of aluminum alloy heating and/or cooling ducts need not be painted, provided that those surfaces have been anodized in accordance with MIL-A-8625, Type II, and sealed Aluminum alloy duct work treated in accordance with MIL-C-5541, Class 1A, shall be painted as required for interior surfaces (see tables I and II) Prior to application of insulation material, the exterior surfaces of insulated duct work, regardless of composition, shall be coated with material capable of withstanding the temperatures and environment of the system Titanium and nickel alloys do not require painting

3.6.23 Reinforced plastic components. Plastic parts reinforced with fibers shall be finished for protection against erosion. Leading edges of fiber-reinforced plastic radomes, antennae, MAD housings, and other components exposed to the air stream, shall be finished with a rain-erosion resistant coating conforming to MIL-C-83231, MIL-C-83445, MIL-C-85322. MIL-C-85322 requires the use of wash primer conforming to MIL-C-8514, applied in accordance with MIL-C-8507. Other methods of protection, such as a boot of suitable erosion and high temperature resistant material, may be used when approved by the procuring activity. In exterior locations, edges shall be sealed with sealant conforming to MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430. Transparent components, whether glass or plastic, shall not be painted. Plastic parts and surfaces, other than those described above, may be painted for color-matching purposes.

3.6.24 Metal leading edges. Exterior surfaces of metallic leading edges exposed to speeds in excess of 500 knots shall be finished with rain erosion-resistant coatings in accordance with MIL-F-18264. Exterior surfaces shall be aerodynamically smooth.

3.6.25 Helicopter rotor-blade leading edges. The finishing system used on the leading edges of helicopter rotor blades shall prevent deterioration of the underlying surfaces and shall be resistant to erosion from rain, sand particles, seaspray and insects. This type of finishing system is not necessary when the edges are made of corrosion-resistant and erosion-resistant material, such as nickel-plated stainless steel)

Source: https://assist.dla.mil -- Downloaded: 202640-17T08:15Z
Check the source to verify that this is the currentversion before use.

3 6 26 Rubber (natural and synthetic). Natural and synthetic rubber shall not be painted, greased, or oiled.

3 7 Requirements for color, insignia, and markings

3.7.1 Exterior color The exterior color of the aerospace weapons system shall be as specified by the procuring activity. The exterior color of Department of the Navy aircraft shall be in accordance with MIL-STD-2161

3.7.2 Interior color The interior color of the aerospace weapons system shall be in accordance with MIL-C-8779.

3.7 3 Insignia and markings Insignias and markings shall be as specified by the procuring activity. Insignias and markings of Department of Navy aircraft shall be in accordance with MIL-STD-2161. All insignias and markings shall be applied over completed exterior surfaces.

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Inspection tests. All coatings and finishes shall be inspected to ensure conformance to the requirements of this specification. When inspection is performed at a contractor's plant, all inspections and tests shall be conducted by the contractor, and the results thereof shall be retained for review as required by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 202610-17T08:15Z
Check the source to verify that this is the current version before use.

4 2 Rejection and retest. Coatings and finishes not conforming to the requirements of this specification shall be rejected Coatings and finishes may be reworked or replaced to correct defects, and resubmitted for acceptance. Full particulars concerning previous rejection and the action taken to correct the defects found originally shall be furnished to the procuring activity.

5 PACKAGING (THIS SECTION IS NOT APPLICABLE TO THIS DOCUMENT)

6. NOTES

6 1 Intended use. This specification covers the methods and materials used for the protective surface treatment and the application of such finishes and protective coatings on aerospace weapons system parts, including spares, and assemblies, such as fuselages, wings, cowls, strut empennage, rotor blades, and floats. This document does not intend to address the application of solid film lubricants as a finishing system. This specification, unless specifically referenced in the item specification or drawing, does not apply to equipment, such as propellers, electrical and electronic equipment, power plants, and instruments, nor to standard contractor purchased accessories, such as electric motors, generators, switches, hydraulic valves, and similar parts

6.2 Definitions. For the purpose of this specification, the following definitions apply.

6.2.1 Exterior surfaces. All surfaces of a weapons system normally exposed to an external environment during flight or on the ground, and all interior surfaces which may become regularly wetted with water or other corrosive fluid shall be considered exterior surfaces. These surfaces include, but are not limited to wheels and landing gear, wheel wells and their fairings, dive brakes, wing flaps, wing-fold areas, battery compartment, and bilge area on aircraft with latrines.

6.2.2 Fayed surface. The term "fayed surface" means that two or more surfaces that are joined or fitted closely or tightly together.

6.2.3 Hull bottom. The "hull bottom" is that surface area of a seaplane fuselage below a line 12 in. (0.31 m) above the full-load water line.

6.2 4 Seaplanes. All aircraft operating wholly, or in part, from water, such as flying boats, airplanes with float-type alighting gear, aircraft with hydro-skis, amphibians, or convertibles are considered to be "seaplanes".

6.2.5 Weapons system. The term weapons system herein includes all types of aircraft and missiles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

6 3 Publications Because of the general nature of this document's requirements, and because the weapons-system-peculiar contractor prepared specifications take precedence over this specification for the duration of the contract, copies of this specification will not ordinarily be distributed to subcontractors. It is the responsibility of the prime contractor to transmit to each subcontractor such details of the finishing procedure as are applicable to his work.

6.4 Consideration of data requirements The following data requirements should be considered when this specification is applied on a contract The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 227.405-70 exempts the requirements for a DD Form 1423

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.1.2	DI-MFFP-81336	Finish specification	---

The above DID's are those cleared as of the date of this specification. The current issue of DOD 5010 12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.5 Subject term (key word) listing.

Corrosion prevention
Finish specification
Protective coatings
Sealants
Surface treatments

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MR
Navy - AS
Air Force - 11

Review activities:
Army - AR, AV, MI, ME
Navy - CG
Air Force - 14, 84, 99

Preparing activity:
Navy - AS

(Project MFFP-0487)

Source: https://assist.dla.mil -- Downloaded: 202610-17T08:15Z
Check the source to verify that this is the currentversion before use.

TABLE I Primer-topcoat system compatibility

Primer Coating	Topcoat					
	MIL-C-22750 6/	MIL-C-46168 3/ 7/	MIL-C-53039 3/ 7/	MIL-C-83286	MIL-C-85285 6/ 7/	TT-P-2756 4/ 7/
MIL-P-23377 2/ 4/ 6/	X	X	X	X	X	1/
MIL-P-53022 2/ 5/ 7/	X	X	X	X	X	1/
MIL-P-53030 2/ 5/ 6/ 8/	X	X	X	X	X	1/
MIL-P-85582 4/ 6/ 8/	X	X	X	X	X	1/
TT-P-2760 3/ 4/ 6/				X	X	1/

1/ TT-P-2756 is a self-priming topcoat Application of a primer coating is not required, with the exception of color number 36495 of FED-STD-595 For infrared reflectance protection, this color requires the use of a primer coating conforming to type II of TT-P-2760, MIL-P-23377, or MIL-P-85582. TT-P-2756 is authorized for use on aluminum, aluminum alloy, and polymer matrix composite structures only TT-P-2756 is compatible with all of the primer coatings listed above If the item to be coated with TT-P-2756 has been preprimed, removal of the primer coating prior to application of TT-P-2756 is not necessary. TT-P-2756 shall be applied to a dry film thickness of 1.7 to 2.3 mils (43 to 58 µm)

2/ MIL-C-46168, MIL-P-53022, MIL-P-53030, MIL-C-53039, and MIL-P-23377, Class 3 of either Type I or II, are not approved for use on U S Air Force airborne weapons systems.

3/ TT-P-2760 is primarily intended for use on aircraft in areas where there is a high degree of structural flexing TT-P-2760 shall be applied to a dry film thickness of 1.5 to 2 0 mils (38 to 51 µm)

4/ These coatings are best suited for aluminum and polymer matrix composite substrates

5/ These coatings are best suited for ferrous and magnesium substrates

6/ Contains at least one type or class with a maximum volatile organic compound content (VOC) of 340 grams/liter (2 8 pounds/gallon)

7/ Contains at least one type or class with a maximum volatile organic compound content (VOC) of 420 grams/liter (3 5 pounds/gallon)

8/ This material may cause flash rusting on bare steel Do not use on bare steel unless proven satisfactory for the intended purpose

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

TABLE II. Protective finish system requirements.

Item	Material	Minimum applied dry film thickness, mils(μm)			
		Primer		Topcoat	
		Exterior	Interior	Exterior	Interior
1	All aluminum alloys (excet bottoms and interior trailing edge control surfaces, for which item 7 applies).	0 6 (15)	1 2 (30)	1 7 (43)	-
2	Sacrificial metal coatings and nonsacrificial coatings applied to non corrosion-resistant metals	0.6 (15)	1 2 (30)	1 7 (43)	-
3	Titanium, alloys 1/	-	-	-	-
4	Magnesium alloys	1 2 (30)	1 2 (30)	1 7 (43)	1 7 (43)
5	Armor plate - ferrous	0 6 (15)	1 2 (30)	1.7 (43)	-
6	Corrosion resistant alloys	0 6 (15)	0 6 (15)	1.7 (43)	-
7	All metals not covered by above	0 9 (23)	1 5 (38)	1 7 (43)	-
8	Polymer matrix composites	0.6 (15)	2/ 1 2 (30)	1.7 (43)	-

1/ These metals do not require primer coating or topcoats for corrosion protection except for faying surfaces as noted in 3 6 2 and 3 6.3 Primer coatings and topcoats may be applied to blend with adjacent areas (use Item 3 requirements).

2/ Application of primer on interior surfaces is only required at dissimilar metal interfaces (see 3 6 2 2c)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.

TABLE III Tubing categories and required coatings

Category	Category description	Primer coating 1/	Final paint system 2/
I	Single tubes having separable connections at each end	Applied after all required forming operations have been completed and prior to fabrication of the assembly	Topcoat applied after fabrication and prior to installation.
II	Assemblies made up of individual tubes permanently joined by nonseparable type fittings (brazing, welding, swaging) and having separable type connectors at each end	Primer coating applied, followed by application of sealant (MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430) after all required bending and permanent joining has been completed and prior to final fabrication of the assembly	Same as for category I
III	Single or multiple tube assemblies having one or more free ends which must be permanently joined by nonseparable type fittings (brazing, welding, swaging) to another tube assembly during installation.	Same as for category II. Additionally, for tube assemblies employing a permanent joining process not compatible with the primer coating during fabrication, the primer coating may be omitted a suitable distance from the affected free ends	Same as for category I Additionally, for assemblies having been only partially primed, additional primer coating shall be applied as required and, when feasible, followed by the coating of all nonseparable joints with sealant (MIL-S-8802, MIL-S-29574, MIL-S-81733, or MIL-S-83430) followed by the required exterior paint system
IV	Other types of tube assemblies not covered in categories I, II, or III. For this category, the contractor shall establish a suitable paint protection system acceptable to the procuring activity.	Not applicable	Not applicable

1/ Primer coatings shall be applied in accordance with tables I and II Assemblies in categories I, II, and III, in which sleeves or ferrules are used in the separable connection, and the sleeves or ferrules are fixed in position by deformation of one or both members into contact, the primer coating need not extend beyond the initial point of intimate contact For all tubing categories where flare fittings are used, primer coating must be applied to the end of the tube

2/ Any damage occurring to the finish system during installation shall be touched up For aluminum plumbing lines, see 3 6 12 1 2

Source: https://assist.dla.mil -- Downloaded: 20260-17T08:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-7179G	930414

3. DOCUMENT TITLE
FINISHES, COATINGS, AND SEALANTS FOR THE PROTECTION OF AEROSPACE WEAPONS SYSTEMS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	b. TELEPHONE (Include Area Code) (1) Commercial: (908) 323-7488 (2) AUTOVON: 624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:15Z
Check the source to verify that this is the current version before use.